

NUMERICAL SOLUTION OF ORDINARY DIFFERENTIAL EQUATIONS

Taylor's series

We consider the differential equation

$$y' = f(x, y)$$

with the initial condition

$$y(x_0) = y_0$$

If $y(x)$ is the exact solution, then the Taylor's series for $y(x)$ around $x = x_0$ is given by

$$y(x) = y_0 + (x - x_0)y'_0 + \frac{(x - x_0)^2}{2!}y''_0 + \dots$$

If the values of $y'_0, y''_0, \dots$ are known, then above eq. gives a power series for y . Using the formula for total derivatives, we can write

$$y'' = f' = f_x + y'f_y = f_x + ff_y,$$

where the suffixes denote partial derivatives with respect to the variable concerned. Similarly, we obtain

$$y''' = f_{xx} + 2ff_{xy} + f^2f_{yy} + f_xf_y + ff_y^2$$

and other higher derivatives of y . The method can easily be extended to simultaneous and higher-order differential equations.

Example: From the Taylor series for $y(x)$, find $y(0.1)$ correct to four decimal places if $y(x)$ satisfies

$$y' = x - y^2 \quad \text{and} \quad y(0) = 1$$

Solution: The Taylor series for $y(x)$ is given by

$$y(x) = 1 + xy'_0 + \frac{x^2}{2}y''_0 + \frac{x^3}{6}y'''_0 + \frac{x^4}{24}y^{iv}_0 + \frac{x^5}{120}y^v_0 + \dots$$

The derivatives $y'_0, y''_0, \dots$ etc. are obtained thus:

$$\begin{aligned} y'(x) &= x - y^2 & y'_0 &= -1 \\ y''(x) &= 1 - 2yy' & y''_0 &= 3 \\ y'''(x) &= -2yy'' - 2y'^2 & y'''_0 &= -8 \\ y^{iv}(x) &= -2yy''' - 6y'y'' & y^{iv}_0 &= 34 \\ y^v(x) &= -2yy^{iv} - 8y'y''' - 6y''^2 & y^v_0 &= -186 \end{aligned}$$

Using these values, the Taylor series becomes

$$y(x) = 1 - x + \frac{3}{2}x^2 - \frac{4}{3}x^3 + \frac{17}{12}x^4 - \frac{31}{20}x^5 + \dots$$

To obtain the value of $y(0.1)$ correct to four decimal places, it is found that the terms up to x^4 should be considered, and we have $y(0.1) = 0.9138$.

Example: Given the differential equation

$$y'' - xy' - y = 0$$

with the conditions $y(0) = 1$ and $y'(0) = 0$, use Taylor's series method to determine the value of $y(0.1)$.

Solution: We have $y(x) = 1$ and $y'(x) = 0$ when $x = 0$. The given differential equation is

$$y''(x) = xy'(x) + y(x)$$

Hence $y''(0) = y(0) = 1$.

$$\begin{aligned} y'''(x) &= xy''(x) + y'(x) + y'(x) = xy''(x) + 2y'(x), \\ y^{iv}(x) &= xy'''(x) + y''(x) + 2y''(x) = xy'''(x) + 3y''(x), \\ y^v(x) &= xy^{iv}(x) + y'''(x) + 3y'''(x) = xy^{iv}(x) + 4y'''(x), \\ y^{vi}(x) &= xy^v(x) + y^{iv}(x) + 4y^{iv}(x) = xy^v(x) + 5y^{iv}(x), \end{aligned}$$

and similarly for higher derivatives. Putting $x = 0$, we obtain $y'''(0) = 2y'(0) = 0$, $y^{iv}(0) = 3y''(0) = 3$, $y^v(0) = 0$, $y^{vi}(0) = 5$.

By Taylor's series, we have

$$y(x) = y(0) + xy'(0) + \frac{x^2}{2}y''(0) + \frac{x^3}{6}y'''(0) + \frac{x^4}{24}y^{iv}(0) + \frac{x^5}{120}y^v(0) + \frac{x^6}{720}y^{vi}(0) + \dots$$

Hence

$$\begin{aligned} y(0.1) &= 1 + \frac{(0.1)^2}{2} + \frac{(0.1)^4}{24}(3) + \frac{(0.1)^6}{720}(5) + \dots \\ &= 1 + 0.005 + 0.0000125, \text{ neglecting the last term} \\ &= 1.0050125, \text{ correct to seven decimal places.} \end{aligned}$$

Picard's Method

$$y = y_0 + \int_{x_0}^x f(x, y) dx$$

Above equation, in which the unknown function y appears under the integral sign, is called an integral equation. Such an equation can be solved by the method of successive approximations.

In which the first approximation to y is

$$y^{(1)} = y_0 + \int_{x_0}^x f(x, y_0) dx$$

second approximation $y^{(2)}$:

$$y^{(2)} = y_0 + \int_{x_0}^x f(x, y^{(1)}) dx$$

Proceeding in this way, we obtain $y^{(3)}, y^{(4)}, \dots, y^{(n-1)}$ and $y^{(n)}$, where

$$y^{(n)} = y_0 + \int_{x_0}^x f(x, y^{(n-1)}) dx \quad \text{with } y^{(0)} = y_0$$

Example: Solve the equation $y' = x + y^2$, subject to the condition $y = 1$ when $x = 0$.

Solution: We start with $y^{(0)} = 1$ and obtain

$$y^{(1)} = 1 + \int_0^x (x + 1) dx = 1 + x + \frac{1}{2}x^2$$

Then the second approximation is

$$\begin{aligned} y^{(2)} &= 1 + \int_0^x \left[x + \left(1 + x + \frac{1}{2}x^2 \right)^2 \right] dx \\ &= 1 + x + \frac{3}{2}x^2 + \frac{2}{3}x^3 + \frac{1}{4}x^4 + \frac{1}{20}x^5 \end{aligned}$$

It is obvious that the integrations might become more and more difficult as we proceed to higher approximations.

Example: Given the differential equation

$$\frac{dy}{dx} = \frac{x^2}{y^2 + 1}$$

with the initial condition $y = 0$ when $x = 0$, use Picard's method to obtain y for $x = 0.25, 0.5$ and 1.0 correct to three decimal places.

Solution: We have

$$y = \int_0^x \frac{x^2}{y^2 + 1} dx$$

Setting $y^{(0)} = 0$, we obtain

$$y^{(1)} = \int_0^x x^2 dx = \frac{1}{3}x^3$$

and

$$y^{(2)} = \int_0^x \frac{x^2}{(1/9)x^6 + 1} dx = \tan^{-1} \left(\frac{1}{3}x^3 \right) = \frac{1}{3}x^3 - \frac{1}{81}x^9 + \dots$$

so that $y^{(1)}$ and $y^{(2)}$ agree to the first term, viz., $(1/3)x^3$. To find the range of values of x so that the series with the term $(1/3)x^3$ alone will give the result correct to three decimal places, we put

$$\frac{1}{81}x^9 \leq 0.0005$$

which yields

$$x \leq 0.7$$

Hence

$$\begin{aligned} y(0.25) &= \frac{1}{3}(0.25)^3 = 0.005 \\ y(0.5) &= \frac{1}{3}(0.5)^3 = 0.042 \\ y(1.0) &= \frac{1}{3} - \frac{1}{81} = 0.321 \end{aligned}$$

Euler's method

$$y_1 = y_0 + \int_{x_0}^{x_1} f(x, y) dx$$

Assuming that $f(x, y) = f(x_0, y_0)$ in $x_0 \leq x \leq x_1$, this gives Euler's formula

$$y_1 \approx y_0 + hf(x_0, y_0)$$

Similarly for the range $x_1 \leq x \leq x_2$, we have

$$y_2 = y_1 + \int_{x_1}^{x_2} f(x, y) dx$$

Substituting $f(x_1, y_1)$ for $f(x, y)$ in $x_1 \leq x \leq x_2$ Proceeding in this way, we obtain the general formula

$$y_{n+1} = y_n + hf(x_n, y_n), \quad n = 0, 1, 2, \dots$$

The process is very slow and to obtain reasonable accuracy with Euler's method, we need to take a smaller value for h . Because of this restriction on h , the method is unsuitable for practical use and a modification of it, known as the modified Euler method, which gives more accurate results.

Example: To illustrate Euler's method, we consider the differential equation $y' = -y$ with the condition $y(0) = 1$ with $h = 0.01$.

Solution:

$$y_{n+1} = y_n + hf(x_n, y_n), \quad n = 0, 1, 2, \dots$$

$$y(0.01) = 1 + 0.1(-1) = 0.99$$

$$y(0.02) = 0.99 + 0.01(-0.99) = 0.9801$$

$$y(0.03) = 0.9801 + 0.01(-0.9801) = 0.9703$$

$$y(0.04) = 0.9703 + 0.01(-0.9703) = 0.960$$

The exact solution is $y = e^{-x}$ and from this the value at $x = 0.04$ is 0.9608.

Modified Euler's method

Instead of approximating $f(x, y)$ by $f(x_0, y_0)$, we now approximate the integral by means of trapezoidal rule to obtain

$$y_1 = y_0 + \frac{h}{2} [f(x_0, y_0) + f(x_1, y_1)]$$

We thus obtain the iteration formula

$$y_1^{(n+1)} = y_0 + \frac{h}{2} \left[f(x_0, y_0) + f(x_1, y_1^{(n)}) \right], \quad n = 0, 1, 2, \dots$$

where $y_1^{(n)}$ is the n th approximation to y_1 . The iteration formula can be started by choosing $y_1^{(0)}$ from Euler's formula:

$$y_1^{(0)} = y_0 + hf(x_0, y_0)$$

Example: Determine the value of y when $x = 0.1$ given that

$$y(0) = 1 \quad \text{and} \quad y' = x^2 + y$$

We take $h = 0.05$. With $x_0 = 0$ and $y_0 = 1.0$, we have $f(x_0, y_0) = 1.0$.

Solution: Hence Euler's formula gives

$$y_1^{(0)} = 1 + 0.05(1) = 1.05$$

Further, $x_1 = 0.05$ and $f(x_1, y_1^{(0)}) = 1.0525$.

The average of $f(x_0, y_0)$ and $f(x_1, y_1^{(0)})$ is 1.0262.

The value of $y_1^{(1)}$ can therefore be computed by using

$$y_1^{(n+1)} = y_0 + \frac{h}{2} \left[f(x_0, y_0) + f(x_1, y_1^{(n)}) \right], \quad n = 0, 1, 2, \dots$$

and we obtain

$$y_1^{(1)} = 1.0513$$

Repeating the procedure, we obtain $y_1^{(2)} = 1.0513$. Hence we take $y_1 = 1.0513$, which is correct to four decimal places.

Next, with $x_1 = 0.05, y_1 = 1.0513$ and $h = 0.05$, we continue the procedure to obtain y_2 , i.e., the value of y when $x = 0.1$. The results are

$$y_2^{(0)} = 1.1040, \quad y_2^{(1)} = 1.1055, \quad y_2^{(2)} = 1.1055$$

Hence we conclude that the value of y when $x = 0.1$ is 1.1055.